

Cem Özçelik

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PROFESSIONAL SUMMARY

Senior Data Scientist with 5 years of end-to-end ML lifecycle experience and an Industrial Engineering foundation - covering the full pipeline from data extraction and feature engineering through statistical modelling, model deployment, and business transfer. Bilingual Turkish-English professional with deep domain expertise in anomaly detection, predictive modelling, and root cause analysis across manufacturing, supply chain, and financial risk. Known for a cost-function-first approach - optimizing for the right loss objective before touching architecture - delivering measurable outcomes: **15% increase in critical machine availability** via predictive maintenance, **97% Recall** on medical diagnostic models, and an **8h/week operational overtime reduction** through ML-driven supply chain forecasting.

CORE COMPETENCIES

Skills: End-to-End ML Lifecycle, Statistical Modelling, Anomaly Detection, Feature Engineering, Predictive Modelling, Data Extraction & ETL Pipelines, Root Cause Analysis

Tools: Python, SQL, Apache Spark / PySpark, XGBoost, Databricks, Docker, Django / FastAPI, Git

WORK EXPERIENCE

Anadolu Pet Akvaryum & Effeffe

Jan 2026 - Present

Senior Database Analyst · Maltepe, Istanbul

- Architected an end-to-end ML deployment platform using Django REST APIs - ingesting raw supply chain data, serving predictive models, and transferring analytical capabilities to business units for data-driven decision making.
- Directed advanced predictive operations using Python-based supply chain forecasting algorithms, sustaining an **8-hour weekly reduction in operational overtime** through ML-driven pipeline automation.
- Translated Association Rule Mining outputs into strategic business actions across SKU portfolios, driving continuous revenue growth through analytical insights.
- Tech:** Python, Django, REST APIs, ML pipelines, Association Rule Mining, PostgreSQL, Docker

Anadolu Pet Akvaryum & Effeffe

Sept 2024 - Dec 2025

IT & Data Analytics Specialist · Maltepe, Istanbul

- Architected and deployed a full-stack internal web application using Python (Django) and SOLID principles - establishing the foundational ML deployment infrastructure from scratch.
- Engineered initial predictive supply chain optimization models using Python and PL/SQL, reducing stockouts and improving inventory turnover through data-driven forecasting.
- Conducted Route-Level Profitability Analysis with advanced SQL, delivering a **3% improvement in overall logistics efficiency**.
- Tech:** Python, Django, PL/SQL, Oracle, SOLID, full-stack deployment

Birinci Automotive

Jan 2022 - Sept 2024

Mid-Level Data Analyst · Çayirova, Kocaeli

- Developed a full predictive ML pipeline - data extraction via MSSQL ETL, feature engineering on machine sensor time-series, unsupervised anomaly detection modelling - delivering a **15% weekly increase in critical machine availability** and **15% OEE improvement**.
- Engineered robust ETL pipelines in MSSQL and Python to ingest and process high-volume manufacturing sensor data, replacing manual workflows with automated, data-driven processes.
- Applied advanced statistical analysis and root cause analysis on complex time-series data for cycle time optimization and operational problem-solving.
- Tech:** Python, MSSQL, ETL pipelines, unsupervised clustering, anomaly detection, time-series analysis

Junior Data Analyst · Çayırova, Kocaeli

- Conducted EDA and applied statistical modelling methods to workforce datasets, uncovering hidden patterns to support data-driven HR policy decisions.
- Designed and implemented automated data ingestion pipelines, **improving reporting efficiency by 15%**.
- **Tech:** Python, SQL, statistical analysis, ETL

PROJECTS

End-to-End Credit Risk Prediction - MLOps & Full Lifecycle Deployment github.com/ccemozclk/Credit-Risk

Full end-to-end ML implementation from raw data extraction to production: feature engineering → XGBoost modelling (63% F1-Score, class imbalance optimized) → interactive Django EDA platform → containerized Docker application. Demonstrates complete ML lifecycle and portfolio/risk analytical thinking applied to financial default detection.

Python · XGBoost · Django · Docker · Feature Engineering · Financial Risk · MLOps

AI-Powered Medical Diagnostics - Explainable AI & CNNs github.com/ccemozclk/Medical-Diagnostics

End-to-end deep learning diagnostic system for pneumonia detection from chest X-rays. Achieved **97% Recall** by optimizing for minimization of False Negatives - cost-function thinking applied to clinical context. Grad-CAM visual heatmaps for model explainability. Transfer Learning on VGG16/ResNet50 for imbalanced medical datasets.

Python · TensorFlow-Keras · CNN · Transfer Learning · Grad-CAM · Explainable AI

Scalable Recommendation Engine - PySpark & Distributed ML github.com/ccemozclk/Spark-Recommendation-Engine

End-to-end personalized recommendation system on Apache Spark - developing strategies by analyzing large data volumes. PySpark + Databricks for distributed processing. K-Means customer segmentation + ALS collaborative filtering for personalized recommendations at scale.

Python · PySpark · Apache Spark · Databricks · K-Means · ALS · Distributed Computing

EDUCATION

Kırıkkale University - B.Sc. Industrial Engineering

2016 - 2021

Rigorous mathematical foundation in statistical analysis, linear algebra, and calculus. MILP optimization with IBM ILOG CPLEX. Multi-Criteria Decision Making (AHP, TOPSIS). Operations research principles directly applied to ML cost-function optimization and analytical thinking.

CERTIFICATIONS

DataCamp - Associate Data Scientist Career Track (PySpark · Feature Engineering · Recommendation Engines) 2024

Miuul - Data Science & Machine Learning Bootcamp (end-to-end ML lifecycle, 10+ real-world projects) 2022

SKILLS

Core Languages: Python (Advanced), SQL (Advanced), Bash

ML & Deep Learning: TensorFlow, PyTorch, XGBoost, Scikit-learn, h2o.ai, CNNs, RNNs, Computer Vision, Explainable AI (Grad-CAM), Transfer Learning

Big Data & Distributed: Apache Spark (PySpark), Databricks, Distributed Computing

MLOps & Deployment: Docker, Django, FastAPI, Flask, REST APIs, Git, Linux, End-to-End ML Deployment

Cloud: AWS (EC2, S3) - GCP-ready

Data Engineering: ETL Pipelines, PostgreSQL, MSSQL, Oracle, Pandas, NumPy

Languages: Turkish (Native), English (Fluent), Spanish (Basic)